NovEdit: A Benchmark of Novice Program Edits for Next-Action Prediction

Jonas Hartmann, Benedicte Ilunga, Rafael Costa

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Abstract

We release NovEdit, 1.2 million edit events from 3,400 consenting novice programmers across four terms, with two benchmark tasks: predicting the next action class (edit, run, delete block, seek help, idle) and predicting whether a student will recover from an error within five minutes. Sequence models outperform count-based baselines on next action (macro-F1 0.58 vs 0.47) but not on recovery (AUC 0.71 vs 0.70). We document collection, consent and splits grouped by student and term.

1 Motivation

Models of novice programming are usually trained on private data from a single course, which makes them hard to compare. NovEdit is a shared benchmark with fixed splits and a datasheet.

2 Collection

Data came from an instrumented editor in an introductory Python course over four terms. Students opted in (consent rate 71%). Events were tokenized, identifiers were hashed per student, and string literals were removed.

Split	Students	Terms	Events
Train	2,380	Terms 1-3	840k
Validation	340	Terms 1-3	118k
Test	680	Term 4	242k

Splits are grouped by student; the test set is a later term to measure temporal shift.

3 Tasks and baselines

Next action: given the last 20 events, predict the next action class. Recovery: given the events up to an error, predict whether the student reaches a passing run within five minutes.

Model	Next action (macro-F1)	Recovery (AUC)
Majority	0.14	0.50
Counts + gradient boosting	0.47	0.70
GRU over events	0.55	0.70
Transformer (6 layers)	0.58	0.71

4 Discussion

Recovery appears to depend on information not present in the edit stream, such as whether a student understood the error message. We encourage submissions that combine edits with other channels,

including assistant use.

5 Responsible use

NovEdit must not be used to identify individual students. We prohibit linking it with other data sources in the license.

References

[1] Gebru, T. et al. Datasheets for datasets. Communications of the ACM 64, 12 (2021), 86-92.

[2] Ferreira, A. and Kowalczyk, P. Group-aware splits for educational sequence data. JEDM 15, 2 (2023), 44-67.